ABSTRACT

Large Language Models (LLMs) are now embedded in chat assistants, document-processing tools, Retrieval-Augmented Generation (RAG) applications, and enterprise automation. Because these systems accept natural-language instructions, they are exposed to prompt injection, jailbreak attacks, malicious instructions hidden in documents, leakage of sensitive information, and other forms of model abuse. Defences based only on static keywords or fixed patterns are easy to bypass and do not keep pace with evolving attacks. This project proposes AI Firewall, a multi-layer security framework that protects an LLM application before and after the model runs. User inputs are examined by a keyword filter, a regular-expression pattern filter, and an on-device AI classifier. A risk scoring engine combines the three results into a low, medium, or high risk level, and high-risk requests are blocked before they reach the LLM.

For RAG applications, uploaded documents are screened before they are added to the retrieval index through text extraction, OCR, hidden-instruction detection, and content cleaning. After the LLM responds, an output firewall checks the response for sensitive information leakage, dangerous instructions, jailbreak indicators, and policy violations. Attacks detected only by the classifier are logged and used to propose new keywords and patterns, which are added to the fast filters only after human review and approval. The proposed framework thus integrates input filtering, secure RAG document screening, semantic classification, risk-based decisions, output filtering, and continuous lexicon learning, and the prototype will be evaluated using detection, latency, document-screening, and output-filtering metrics.